FITNESSBUDDY

AI-Powered Conversational Fitness Coach

PROJECT REPORT

Problem Statement: No. 13 – Fitness Buddy

Prepared by: Mamathasri

Programme: B.Tech – Computer Science and Engineering, AI & ML Specialization

Institution: Vel Tech University, Chennai

Project Summary

FitnessBuddy is a conversational, AI-powered health and fitness assistant. It is designed to provide accessible, friendly and on-demand guidance for people who want to build healthier routines. The assistant can recommend home workouts based on user input, share motivational tips, suggest simple nutritious meal ideas and encourage habit-building and consistency. The project uses IBM Granite through IBM watsonx.ai as its mandatory AI technology, with Python and Gradio for the application and user interface.

1. Introduction

Maintaining a healthy lifestyle can be challenging because of busy schedules, limited time, lack of personalised guidance and inconsistent motivation. Some traditional fitness options require paid subscriptions, in-person consultations or fixed schedules. A conversational virtual assistant can make basic fitness information easier to access and help users explore practical ideas that fit their daily routines.

FitnessBuddy addresses this need through a simple chat-based interface. Users can ask questions about home workouts, healthy meal ideas, motivation and daily habits. The AI assistant responds in clear and supportive language.

2. Problem Statement

Many individuals struggle to maintain a healthy lifestyle due to a lack of personalised guidance, time constraints and inconsistent motivation. There is a need for an accessible, friendly and intelligent virtual assistant that provides on-demand fitness advice, healthy-lifestyle suggestions and basic nutrition guidance tailored to individual needs and available at any time.

3. Objectives

• Recommend home workouts and routines based on user input.
• Provide motivational tips and daily fitness inspiration.
• Suggest simple, nutritious meal ideas.
• Encourage habit-building and consistency.
• Provide an easy-to-use conversational interface.
• Demonstrate the integration of IBM Granite into a practical student application.

4. Proposed System

FitnessBuddy provides a chat interface where the user types a request. The Python application prepares a prompt containing the assistant's role and relevant conversation context. It sends the prompt to an IBM Granite model through IBM watsonx.ai. The generated answer is returned to the application and displayed to the user through Gradio.

5. Main Features

Home workout recommendations: Gives general home-workout ideas based on the user's stated goals, available time, experience and equipment.
Motivational support: Shares encouraging messages and daily fitness inspiration.
Nutritious meal ideas: Suggests simple meal ideas based on food preferences.
Habit-building: Encourages realistic routines and consistency rather than extreme changes.
Conversational interaction: Allows users to ask follow-up questions naturally.

6. Technology Stack

Python: Implements the application logic and prepares requests for the AI model.
Gradio: Creates the web-based chat interface.
IBM watsonx.ai: Provides access to the hosted foundation model service.
IBM Granite: Acts as the generative AI model that creates conversational responses.
GitHub: Stores and shares the project source code and documentation.

7. System Workflow

1. The user opens the FitnessBuddy application.
2. The user enters a fitness-related question or request.
3. Gradio passes the message and chat history to the Python function.
4. Python prepares the prompt and sends it to IBM watsonx.ai.
5. IBM Granite processes the prompt and generates a response.
6. The response is returned to the Python application.
7. Gradio displays the answer to the user.

8. System Architecture

User Fitness question / preferences
Gradio Chat Interface Receives user messages and displays answers
Python Application Prepares prompt and manages conversation
IBM watsonx.ai API Connects the app to the model service
IBM Granite Model Generates conversational fitness guidance
Response to User Workout ideas • Motivation • Meal ideas • Habits

9. Implementation Overview

The application is implemented in Python. Gradio provides the chat interface and collects IBM Cloud API key, watsonx.ai project ID and region URL from the user at runtime. The Python function creates a model connection using the IBM watsonx.ai SDK, builds a prompt and requests a response from the selected IBM Granite model. The answer is displayed in the chat interface.

The supplied starter code is configured with the model ID **ibm/granite-3-3-8b-instruct**. Model availability can vary by IBM Cloud region and account. If this model is not available, select a currently supported IBM Granite model in watsonx.ai and update the model ID in the code.

10. How to Run the Project

Install dependencies:

```
pip install -r requirements.txt
```

Run the application:

```
python app.py
```

Open the local Gradio URL displayed in the terminal. Enter your IBM Cloud API key, watsonx.ai project ID and the region where your project is hosted. Then start chatting with FitnessBuddy.

11. GitHub Repository Contents

Repository: https://github.com/Mamathasri99/Fitness-buddy

The repository should contain:
• app.py
• README.md
• requirements.txt
• problem_statement.pdf
• project_presentation.pptx
• .gitignore
• Other required project files

12. Expected Outcome

The expected outcome is a working prototype of a conversational AI fitness assistant. It should accept natural-language questions and return relevant general workout suggestions, motivational tips, nutritious meal ideas and habit-building guidance.

13. Limitations and Responsible Use

FitnessBuddy is a student prototype that provides general educational information. It is not a doctor, registered dietitian or qualified personal trainer. It cannot diagnose or treat medical conditions, and it cannot guarantee weight, strength or health outcomes. AI-generated content may be incomplete or inaccurate and should be reviewed carefully. Users with pain, injury, pregnancy, chronic conditions or other health concerns should consult a qualified healthcare professional before changing their exercise or diet routine.

14. Future Enhancements

- Save user preferences and previous conversations securely.
- Add optional workout reminders and habit check-ins.
- Provide progress-tracking charts.
- Add multilingual conversation support.
- Improve personalisation while protecting user privacy.
- Add carefully reviewed exercise demonstrations and accessibility improvements.

15. Conclusion

FitnessBuddy demonstrates how conversational AI can be used to make basic fitness and healthy-lifestyle guidance more accessible. By combining Python, Gradio, IBM watsonx.ai and IBM Granite, the project aims to provide a friendly assistant for home-workout ideas, motivation, meal suggestions and habit-building. The project also gives the student practical experience in integrating a foundation model into an application.

Note for submission: *Test the application with your own IBM watsonx.ai project and confirm the model ID and region before claiming that the live integration is working. Never upload API keys or other credentials to GitHub.*